

IOWA CITY, IA, USA

WMO: 725462

Lat: 41.633N

Lon: 91.543W

Elev: 650

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 14937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.5	(c) 0.6	(d) -11.8	(e) 3.0	(f) -3.8	(g) -7.7	(h) 3.7	(i) 1.6	(j) 26.7	(k) 25.3	(l) 24.6	(m) 22.6	(n) 7.4	(o) 310	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.7	92.0	76.2	89.6	75.3	86.7	73.6	79.5	87.6	77.7	86.0	75.9	84.0	9.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	144.6	85.1	75.0	134.7	82.9	73.2	126.3	80.8	43.5	87.9	41.5	85.8	39.8	84.1	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.6	(b) 20.1	(c) 18.0	(e) -12.9	(f) 96.0	(g) 6.1	(h) 3.5	(i) -17.3	(j) 98.5	(k) -20.9	(l) 100.5	(m) -24.3	(n) 102.5	(o) -28.8	(p) 105.0		
(4) 23.6	(5) 20.1	(6) 18.0	(7) -13.4	(8) 82.7	(9) 6.0	(10) 1.2	(11) -17.7	(12) 83.6	(13) -21.2	(14) 84.3	(15) -24.5	(16) 85.0	(17) -28.9	(18) 85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	22.8	26.1	38.6	51.0	62.0	71.5	75.2	72.8	65.9	52.7	40.0	27.3	(6)
(7)		DBStd	20.65	11.89	12.00	11.38	9.38	8.39	6.16	5.71	5.13	7.72	9.39	9.99	11.49	(7)
(8)		HDD50	3120	844	671	386	98	8	0	0	0	1	79	326	707	(8)
(9)		HDD65	6224	1309	1089	821	427	159	16	2	5	82	395	750	1169	(9)
(10)		CDD50	3345	0	2	32	128	381	644	780	706	479	164	26	3	(10)
(11)		CDD65	973	0	0	2	7	67	210	317	246	109	15	0	0	(11)
(12)		CDH74	9049	0	0	14	122	631	1910	3117	2044	1046	163	2	0	(12)
(13)		CDH80	3115	0	0	1	23	163	659	1211	669	352	37	0	0	(13)
(14)	Wind	WSAvg	7.7	8.8	8.7	9.2	9.5	8.3	7.0	6.0	5.6	6.1	7.3	8.2	8.1	(14)
(15)	Precipitation	PrecAvg	37.2	1.1	1.3	2.2	3.8	4.7	5.7	4.5	4.3	3.0	2.7	2.2	1.6	(15)
(16)		PrecMax	59.9	2.1	3.9	5.4	8.3	12.4	11.4	13.8	11.6	7.0	7.4	6.9	3.8	(16)
(17)		PrecMin	23.3	0.4	0.4	0.5	1.5	0.6	1.5	0.6	0.2	0.9	0.5	0.2	0.5	(17)
(18)		PrecStd	8.1	0.6	0.8	1.4	1.7	2.3	2.7	3.0	2.7	1.6	1.7	1.7	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.4	62.3	75.6	83.7	89.8	93.1	97.3	94.1	91.9	85.4	72.3	61.1	(19)
(20)		MCWB	45.4	52.6	62.7	65.0	72.7	76.0	78.5	75.2	73.6	69.6	58.6	56.2	(20)	
(21)		2%	DB	46.5	53.8	68.5	77.5	84.1	90.1	92.7	89.9	88.1	78.9	65.5	53.7	(21)
(22)		MCWB	41.6	46.6	57.0	61.7	68.9	74.6	78.3	76.0	72.7	65.4	55.9	49.9	(22)	
(23)		5%	DB	41.8	46.9	62.7	72.6	80.9	87.1	89.8	86.4	84.0	73.4	60.6	47.7	(23)
(24)		MCWB	37.7	42.4	54.6	58.7	66.9	73.1	76.8	74.2	71.0	62.8	52.5	43.9	(24)	
(25)		10%	DB	37.4	42.3	56.0	67.8	77.3	83.8	86.5	83.6	80.5	68.9	55.4	42.5	(25)
(26)		MCWB	34.8	38.5	48.3	55.7	65.0	71.4	74.9	72.7	68.0	59.9	49.0	39.0	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	54.5	64.3	67.1	74.8	80.1	82.5	81.1	77.6	72.0	61.6	58.1	(27)
(28)		MDCB	50.9	60.6	73.8	78.9	85.2	88.0	91.5	89.1	87.2	81.7	68.8	60.9	(28)	
(29)		2%	WB	42.0	47.7	59.8	63.7	71.8	77.7	80.3	78.3	75.1	68.2	57.5	50.0	(29)
(30)		MDCB	46.8	52.8	65.7	73.0	81.3	86.4	88.9	85.7	84.2	75.0	63.8	52.5	(30)	
(31)		5%	WB	38.1	42.5	55.0	60.7	69.4	75.5	78.3	76.4	72.8	64.9	54.0	43.9	(31)
(32)		MDCB	41.3	47.1	61.8	70.6	77.4	84.1	86.7	83.6	80.5	70.7	59.7	46.9	(32)	
(33)		10%	WB	35.2	38.3	49.1	57.5	67.1	73.5	76.5	74.6	70.6	61.3	49.5	38.8	(33)
(34)		MDCB	37.6	41.6	56.0	65.9	74.7	81.0	84.7	81.3	77.8	68.2	55.1	42.0	(34)	
(35)	Mean Daily Temperature Range	MDBR	15.6	16.5	18.5	21.0	20.3	19.1	18.7	19.0	22.3	20.7	18.2	15.3	(35)	
(36)		5% DB	MCDBR	21.5	23.6	27.4	29.3	24.2	22.4	21.2	22.2	25.3	27.0	25.4	21.9	(36)
(37)		MCWBR	16.7	16.9	17.1	16.4	12.5	10.7	9.6	10.7	11.9	14.8	16.9	17.2	(37)	
(38)		5% WB	MCDBR	19.3	22.6	24.0	24.6	21.2	19.4	19.1	18.9	21.1	21.7	23.5	19.9	(38)
(39)		MCWBR	15.4	16.9	16.1	15.7	12.6	11.2	10.3	10.8	11.2	14.4	17.1	17.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.306	0.338	0.384	0.425	0.426	0.442	0.422	0.391	0.343	0.302	0.290	(40)	
(41)		taud	2.404	2.373	2.378	2.264	2.213	2.263	2.204	2.267	2.316	2.450	2.491	2.429	(41)	
(42)		Ebn at Noon	273	286	288	281	271	270	264	266	266	269	268	263	(42)	
(43)		Edh at Noon	25	30	34	41	45	43	45	41	36	28	23	23	(43)	
(44)	All-Sky Solar Radiation	RadAvg	576	891	1197	1493	1758	1953	1956	1719	1453	959	643	470	(44)	
(45)		RadStd	66	76	105	119	125	188	137	107	115	109	53	53	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page